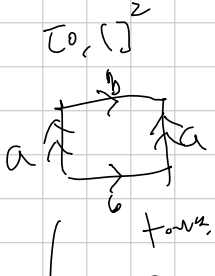
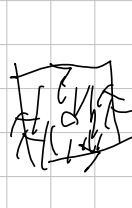


$S^n, \cancel{S^1}, \mathbb{R}P^n, \text{---}$
 $(n \geq 1)$
 $\downarrow$
 $\pi_1 = 0$

2-dim mfd
 $L(p,q)$

$M \cup N$ $\wedge \vee$ $\langle S, S' \rangle / \langle e^{2\pi i}, e^{2\pi i} \rangle$
 τ_0, τ_1


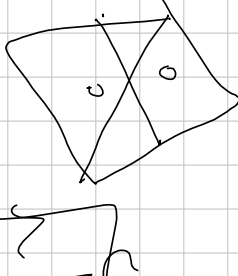
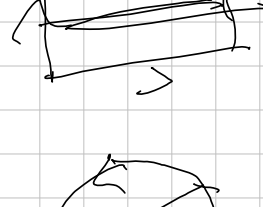
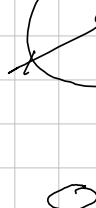
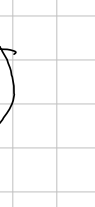
 $\mathbb{Z} \times \mathbb{Z}$ $\langle a, b \mid aba = b \rangle$ free product. $ababab \dots$

$aba^{-1}b^{-1} = e$
 $ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$


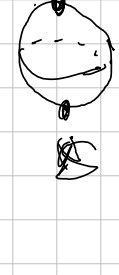



 $S^1 \vee S^1$ $\frac{S^1 \vee S^1}{\mathbb{Z} \times \mathbb{Z}}$ $T: \mathbb{Z} \times \mathbb{Z}$ $T \in \mathbb{Z} \times \mathbb{Z}$ $\mathbb{Z} \times \mathbb{Z}$

Theorem
 $M: n\text{-dim}$
 with $(n \geq 2)$
 then $\pi_1(M) \cong \pi_1(M \setminus \{x\})$
 $\pi_1(\mathbb{R}P^n) \cong \mathbb{Z} \times \mathbb{Z}$ $(n \geq 2)$ $\pi_1(M \setminus \{x\}) \cong \pi_1(M \setminus \{x\})$

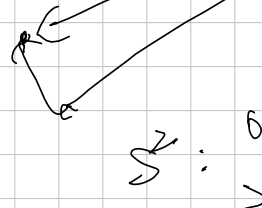






 $\mathbb{Z} \times \mathbb{Z}$ $\frac{S^1 \vee S^1}{\mathbb{Z} \times \mathbb{Z}}$

Example:




 $S^2/p = S$ $\pi_1(\dots)$ $S^1 \vee S^1$



 $S^1: 0\text{-cell } 1\pi$
 $1\text{-cell } 1\pi$
 $S^2: 0\text{-cell } 1\pi$
 $2\text{-cell } 1\pi$
 $\mathbb{R}P^2: 0\text{-cell: } 1\pi$
 $1\text{-cell: } 1\pi$
 $2\text{-cell } 1\pi$

$\mathbb{R}P^2: \mathbb{Z}_2$ $\langle a \mid a^2 \rangle$



